《计算机网络》课程作业（2）

（2月23日布置；3月3日课前手写提交）

1. 什么是RS-232？请简述RS-232的电气特性。

2. Draw the 7-bit waveform diagram that results when the word `Bit` (capital `B`, small letters `i`, `t`) is sent in ASCII across an RS-232 connection.

   (*Hint: to get the 7-bit ASCII character codes, please search "RFC 20" in Baidu for the original document.*)

3. When two humans hold a conversation, do they use simplex, half-duplex, or full-duplex transmission?

4. Which type (or types) of serial transmission is appropriate for video transmission? For a keyboard connection to a computer?

5. Suppose one sent 10,000 7-bit characters across an RS-232 connection that operated at 38,400 baud. How long would the transmission require?

   (*Hint: remember to allow for a start bit and a stop bit on each character.*)

6. Most RS-232 hardware allows the computer to which it attaches to specify the data rate to use as well as the number of stop bits to use. If a transmitter is programmed to use two stop bits, but a receiver is

programmed to require only one, will data be received correctly? If so, what is the disadvantage of using an extra stop bit?